

Developing an advanced machine learning model for identifying virtual montages in digital media

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ARTICLE INFO

Keywords:

Virtual montages

Digital media

Convolutional neural networks (CNN)

Adaptive flower pollination optimization (AFPO)

Mutual information (MI)

Naïve bayes (NB)

ABSTRACT

Virtual montages are compositions that blend several digital materials to produce a new, visually appealing narrative and are becoming increasingly popular, as a result of the quick development of digital media. Virtual montages have emerged as a potent technique for improving visual storytelling and content presentation across platforms in the digital age. However, the quick expansion of digital content has made advanced methods for effectively recognizing and classifying these montages necessary. This work suggests a novel method for virtual montage identification across a broad range of visual themes, content types, and media resolutions, and resolutions utilizing an Adaptive Flower Pollination Optimized Mutual Information with Naïve Bayes (AFPO-MI-NB) model. The research's dataset comprises a variety of content from virtual montage detection dataset which enables the model to generalize effectively to data from the real world. To improve the model's capacity to handle various image and video qualities, pre-processing methods, including data augmentation and pixel value normalization are used. Convolutional neural networks (CNN) are used for feature extraction to capture spatial patterns. The model uses AFPO-MI-NB classification to enhance classification accuracy and computational efficiency. AFPO allows for more adaptable feature weights, MI evaluates the relationship between visual features and the classification task, and NB classifier processes feature matrices for binary classification. The hybrid approach strengthens feature selection through global and local searches, resulting in a model that enhances classification accuracy and improves computational efficiency. According to the experimental data, this model provides a reliable solution for virtual montage identification across a variety of media formats, outperforming current techniques in terms of F1-score of 0.95, recall of 0.92, and precision of 0.97. This work has significant applications in automated digital media analysis, copyright enforcement, and content control.

1. Introduction

A montage, an architectural approach consuming numerous fragments to yield an entire thing, is vital to digital communication. Twitter posts, blog entries, hyperlinks, multimodal fictitious, and revolutionary works often participate in written, visual, and audio characteristics [1]. The prompt spread of information technology over the last thirty years has shaped numerous dissimilar classes that are called virtual montages. It demands efficiently relating and rearranging the digital content to create new, important formations [2]. Montage is an advanced method for high quantity identification of replica information. It makes use of many hereditary bases of information to recognize complex consistencies [3].

Montage identifications perform a vital character in diverse fields, such as criminal investigations, media and entertainment, medical

imaging, patent security, and political domains [4,5]. Present montage technologies carry together and estimate, large amounts of data from multiple resources and they improve technical research by allowing exact imaging of components, consideration of information, and making choices across various multiple sectors [6]. A progressive technique that comprises artificial intelligence (AI) can remarkably raise the montage analysis and deliver actual consequences by proactively identifying inexactness and handling virtual content [7]. The use of cutting-edge techniques stability growth with protections to reduce probable threats and assure the appropriate request of the gadgets [8]. Consuming innovative methods can make the discovery process easier, thereby making it more efficient and expectable even if it has difficult tasks [9].

False reporting, misinformation, digital kidnaps, and malware were typical crimes that used altered digital images and films. Interacted audio and video material were commonly used for multiple online

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<https://doi.org/10.1016/j.entcom.2025.101002>

Received 5 December 2024; Received in revised form 5 May 2025; Accepted 20 July 2025

Available online 22 July 2025

1875-9521/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

criminal activities was described [10]. Automated the identification of altered material and supports law enforcement investigators in the work. The research described a machine learning (ML) model-based montage detection for competitive outcomes and decreased time to process allowed to be perfect for incorporation into self-monitor modules. It estimated the level of falsehood for each handled multimodal file.

Montage analysis in the medical field, particularly for ocular therapies was uses [11]. Diabetes-induced retinopathy (DR) was an important arterial side effect of diabetes, a condition, and visual processing of images with montages played an essential role in identifying, evaluating, and keeping track of DR. Recent developments in ultra-widefield (UWF) scanning had improved visual accessibility, leading to improved diagnostic precision. The technique enhanced productivity by offering medical professionals' greater resources for identifying and managing DR, which contributed to improved outcomes for patients through more efficient therapy and tracking measurements.

In recent days, there have been too many methods available for altering images online, and it is an important concern in forensic investigated [12]. An innovative ML approach to autonomously identify data gathered from an image and correlated the metadata information to existing graphical data based on montages in virtual images was delivered. Metadata could reveal details regarding a picture's origin. The outcomes demonstrated that verification experts could quickly and effectively identify miscellaneous information and falsified metadata during a criminal investigation.

To detect fake or misleading images in measurements, an ML approach is suggested [13]. The combination of hash features made the suitable solution for image montage detection. The outcomes showed improved robustness against replicating. Additionally, the amount of memory needed decreases as only a small amount of the hard drive storage and primary storage will be needed. Overall, the system for recognizing images grew incredibly correctly, even when dealing with an infinite number of photos.

The application of blockchain technology and decentralized transaction technologies (DLTs) as possible ways to tackle digital fraud, particularly deepfakes, disinformation, and misleading information examined [14]. It implied an investigation of the procedures employed by DLTs to provide authenticity and provenance of data stored online. The results offered that blockchain had enormous potential for resolving concerns with the reliability and authenticity of online content. However, challenges remain in regulation, system connectivity, and the widespread adoption of blockchain technologies across industries.

A method using current techniques to detect video forgeries, including computer vision (CV), ML algorithms, and deep learning (DL) approaches of deepfake detection, pixel-based detection, and inter and intra-detection was determined [15]. The outcomes confirmed that the recommended techniques achieved great accuracy and efficiency in classifying different kinds of video alterations. The discoveries highlighted the possibility of linking progressive techniques for strong video fake detection.

With the emerging technology, digital descriptions and videos were operated for violent use. The recommended technique [16] was a new analytical method planned to discover later handling of digital video in the resulting structure of MP MOV format, and 3GP file formats. The method used unsupervised neural networks to detect the explicit communication situation or adapt software that prompts the behavioural modification. The system's outstanding exactness and flexibility for routine use were confirmed by the outcomes of the experiment, which confirmed that it could truthfully detect the social media site or changing software utilized.

False newscasts had comprehensive influences on society, manipulating community opinion, business power, and the possibility for misappropriation in fields such as politics, the law, and remote life. The profound misrepresentation of consuming AI knowledge from an array of angles, such as the consequence of media on society, the purpose of media formation and illustration in the press, the reaction from

audiences, and the larger governmental and lawful inferences of fake information deployment examined [17]. The outcomes of the suggested approach illustrated efficiently that public policy and legislation typically were unprepared for managing the implications of deepfakes. Additionally, the results focused on the importance of apparent shortcomings in the fakes that are currently in use.

The challenge of extremely fine classification of sentiment in online content was investigated [18]. It concentrated on recognizing and assessing the unique semantics of jargon, humour, and emoticons, which influence emotional perception. Utilizing deep learning and neural networks, the suggested approach had been optimized for sentiment detection on a limited number of analyses of the social media platform data. The findings showed better sentiment classification in online communication platforms and precisely forecasted the extremes of viewpoint by capturing the timing consequences of spoken language, laughter, and symbols. The approach exhibited the effectiveness of the model in absorbing and classifying emotion quickly in digital platform settings.

The difficulties of understanding and developing an automated copyright system on social media platforms was analysed [19]. It suggested a classifier that uses an AI technique built on the AI platform language framework that classified videos on YouTube into sub-categories related to documented copyright problems utilizing an inventory of 76.7 million videos. The examination highlighted the variations in adherence across different material groups accurately while providing facts about eliminating percentages. The findings indicated that different levels of computerization influenced content identification and DMCA systems react to copyright violations.

Computer vision and ML approaches to develop an innovative video feature framework. From a content-driven viewpoint, the structure aimed to help advertisers predict and understand video on internet usage used [20]. The findings of the research indicated the structure's ability to improve advanced management and marketing research by making it possible to utilize unorganized footage, generating findings in several applications, including entertaining statistics and video advertising. Also, it emphasized the essential nature of predicting customer behaviour at the user level, including whether viewers complete videos and whether they watch the next video in a series.

To prevent the dissemination of misinformation in society, offered a technique for the verification of the true level of the information [21]. The suggested ML approach was performed in two stages. It showed a need for the security process. Utilizing the language feature for monitoring the written format structures and combining with the categorization of points improved the efficiency of the model. The outcome of the approach demonstrated that the ML approach outperformed the recent DL approaches in terms of effectiveness in the safeguard manner.

Delivered two ML algorithms for detecting news reliability from online network settings was described [22]. Employing the two distinct models on the same gathered information led to the deeper knowledge of the problem scenario. Findings showed that two methods performed effectively for detecting reliable and unreliable information. Finally, the outcomes proved that one ML method had a superior performance than the other.

1.1. Aim of the research

The objective of this research is to improve an innovative method for distinguishing and recognizing digital montages in a variety of multimedia formats, types, and determinations. It deals with the growing contest of spotting influenced internet data by improving machine learning (ML) for identifying virtual montages in news sources, movies, social media, and marketers.

1.2. Key contributions

- To delivered a modern approach that works with the Adaptive Flower Pollination Optimized Mutual Information with Naïve Bayes (AFPO-MI-NB) idea to identify virtual arrangements across a wide range of technical data.
- Utilized state-of-the-art preprocessing methods (data augmentation and pixel normalization) and CNNs to robustly extract spatial features, enabling the model to better generalize across images/videos of varying quality.
- To display that the outcomes of AFPO-MI-NB model hits earlier methods in terms of accuracy and efficiency, creating it a trustworthy selection for discovering virtual montages.

1.3. System organization

This research is structured as follows: Part 2 offers the related work in the field of virtual montage detection, while Part 3 focuses on the methodology used model and its presentation for identifying montages across diverse media types. Part 4 offers the experimental results and performance evaluations, and Part 5 completes the research with discussions on its implementation and possible future suggestions.

2. Research methodology

This section comprises the employed dataset, pre-processing techniques, and feature extraction for identifying the montages using AFPO-MI-NB, and an effective combination process is explained. Fig. 1 shows the entire process flow of the AFPO-MI-NB method.

2.1. Dataset collection

In this research, utilized from the Kaggle dataset (Virtual Montage Detection Dataset) numerous individual frames from videos that are included in the UCF-101 dataset comprise the dataset, a popular computer vision benchmark for action detection. In the initial dataset, every video matched one of 101 human action categories, including archery, guitar playing, basketball dunks, etc. Instead of analyzing the entire movie, this modified version supports image-based modeling by extracting frames. (<https://www.kaggle.com/datasets/programmer3/virtual-montage-detection-dataset>).

2.2. Pre-processing

Before the AFPO-MI-NB model development, the normalization process and augmentation phases are employed in the pre-processing stage that increases the robustness across distinct data.

Min-max normalization: To make the standardization of gathered information and improve the model's ability, pixel value normalization is used. Particularly the min-max normalization is the most suitable for

normalization of digital contents. Min-max normalization is a method that creates equilibrium by comparing data values before and after the process, applying linear transformations to the original data. By ensuring the pixel values are scaled in the specific range, this technique proved the suitability of an advanced method for effective montage identification. Eq. (1) describes the standardization process.

$$y_{new} = \frac{y - \min(y)}{\max(y) - \min(y)} \quad (1)$$

where y_{new} represents the fresh value obtained from the outcomes. y denotes the earlier value $\min(y)$ presents the least value in the dataset and $\max(y)$ offers at most value in the dataset.

Data augmentation: In the framework of advanced techniques for montage identification, data augmentation is crucial because it forces a system to acquire persistent characteristics, assisting in generalization. Recognition of images, multi-media identification, and semantic division are effectively used in vision applications that employ data enhancement. It can offer fluctuations in data and increased capacity in the dataset to increase identification performance. Using augmentation of data as a pre-processing technique can improve image or video quality. It enhances the accuracy of recognizing gathered information. Basic image changes were chosen when the improved quality was needed. It is primarily used for horizontal flips and cropping at random to evaluate the multi-media dataset. It also provides geometric transformations, including cropping, rotating, and flipping, to measure the changed orientations in images. The lower quality information is dropped out. In addition, the color adjustments make it suitable for practical applications, which enhances the model flexibility across varied media designs and qualities.

2.3. Feature extraction

A basic convolutional neural network (CNN) has been suggested in this section to extract the most notable features from the digital information. CNN is the commonly employed technique in artificial intelligence (AI) for extracting complex image data, such as recognizing computer-generated montages, due to its special capability to seize spatial relationships in visual content as displayed in Fig. 2. It is made up of a variety of layers, which utilizes the information that collects it while performing specific tasks. To reduce computational load and enhance the recognition across various image types and manipulations, including virtual montages, the CNN in the application layer is optimized. It is challenging to correctly categorize binary results like montage or non-montage since CNN algorithms establish an enormous amount of complex data for each linked layer. To accomplish better categorization in the AFPO-MI-NB framework, methods for reducing dimensionality using CNN are utilized by transforming the large space of features into more effective representations with fewer variables.

The model is composed of multiple layers intended for montage classification. The input layer takes RGB images of 224x224 pixels (224

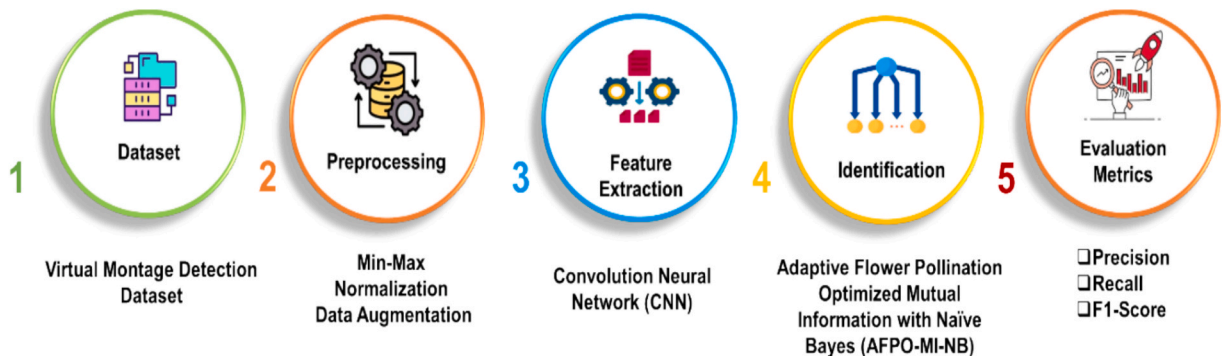


Fig. 1. AFPO-MI-NB process flow.

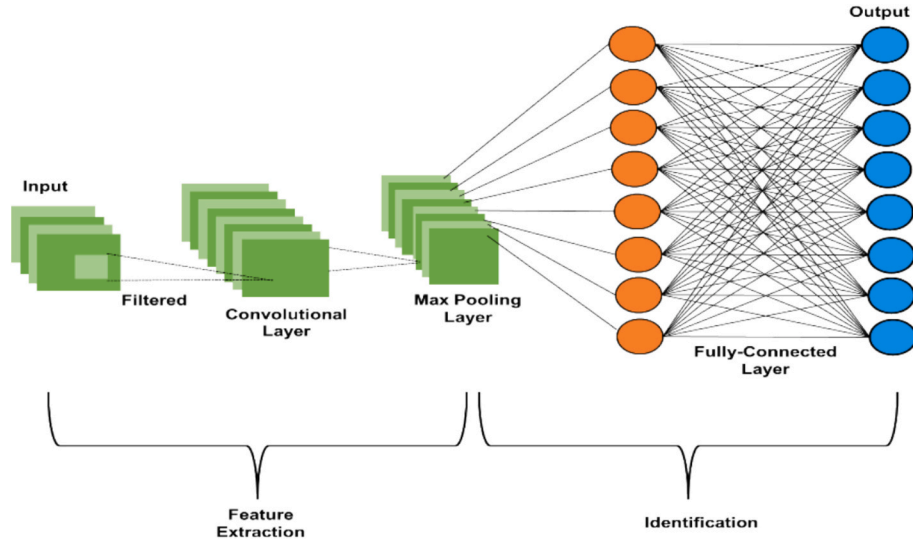


Fig. 2. Feature extraction of CNN.

x 224 x 3) where the images are pre-processed and normalized. The first convolutional layer employs 32 filters with a kernel size of 3×3 , a stride of 1, and 'same' padding to maintain spatial dimensions followed by a ReLU activation function to add non-linearity. The initial max-pooling layer utilizes a window of 2×2 and a stride of 2 to decrease spatial dimensions, reducing computational load while preserving key features. The output of the max-pooling layer is flattened into a 1D array to be processed by fully connected layers. The initial fully connected layer has 128 neurons with ReLU activation, combining the features of the convolutional and pooling layers to learn high-level representations. The output layer, which consists of one neuron and a Sigmoid activation function, makes a prediction of a probability score for binary classification, indicating whether the image is a montage (1) or a non-montage (0).

2.4. Adaptive flower pollination optimized mutual information with Naïve Bayes (AFPO-MI-NB) for classifying virtual montages in digital media

AFPO-MI-NB successfully detects the montage classification across many kinds of data by integrating these optimized characteristics into NB, a model prominent for its reliable categorization through multiple data formats in digital media. This hybrid model targets the importance of montage identification by utilizing the AFPO optimization capacity to dynamically pick the most appropriate aspects. Mutual Information (MI) gives a good approach to the selection of appropriate features, utilizing AFPO, each feature is dynamically weighted based on its importance so that better classification performance can be achieved. The method is innovative since classical methods of feature selection do not involve dynamic feature weighting in the course of classification.

2.4.1. Naïve bayes (NB) for efficient classification

The suitable ML classification approach referred to as the NB classifier was based on a quantitative methodology that makes use of Bayes' principle of probabilities. Min-max normalization is a method that creates equilibrium by comparing data values before and after the process, applying linear transformations to the original data. This represents the main argument for using this algorithm to identify montages from unstructured data based on various types of features. Possibility concerns has become extremely successful because they are unaffected by the existence of the characteristics extracted from many domains that deal with occurrences under variability. To determine whether a sample image is ordinary or comprises a virtual montage, the NB classifier is trained using the features that were taken from the dataset. To increase

overall performance and the classifier's capacity to detect small variations, more false montage was used in the training process of virtual montages can exist in several formats and settings.

The possibly desired value, d_{MAP} , has been assigned, the attributes with values $< b_1, b_2, \dots, b_m >$ that identify the particular instance and - classify the new instance utilizing the Bayesian approach in Eq. (2).

$$d_{MAP} = \underset{d_i \in D}{\operatorname{argmax}} (O(d_i \setminus b_1, b_2, \dots, b_m)) \quad (2)$$

By the application of theories of Bayes given in Eq. (3),

$$\begin{aligned} d_{MAP} &= \underset{d_i \in D}{\operatorname{argmax}} \left(\frac{O(b_1, b_2, \dots, b_m \setminus d_i) O(d_i)}{O(b_1, b_2, \dots, b_m)} \right) \\ &= \underset{d_i \in D}{\operatorname{argmax}} O(b_1 \setminus b_2, b_2, \dots, b_m \setminus d_i) O(d_i) \end{aligned} \quad (3)$$

More standardization can be made practicable through Eq. (4), which infers that, given the target value, the attribute values do not relate to each other.

$$d_{NB} = \underset{d_i \in D}{\operatorname{argmax}} (O(d_i) \prod_j O(b_j \setminus d_i)) \quad (4)$$

Where d_{NB} is the output of the classifier. $b_j \setminus d_i$ is measured with the experimental information. The $O(d_i)$ denotes the earlier possibilities to update in a relevant manner. $(b_j \setminus d_i)$ is the constrained possibility computed from the development model. Eq. (5) offers the standard distribution of the probability density function.

$$e(w) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(w-\mu)^2}{2\sigma^2}} \quad (5)$$

where μ is the average and σ is the variance. This approach makes use of NB due to its simplicity of usage, speed, and effectiveness when dealing with data that is highly dimensional, especially when integrate with methods for selecting features that are monitored. NB is highly efficient, especially when it comes to distinguishing between binary classes like montage and non-montage. It achieves by estimating the probability of every group utilizing the characteristics that were collected, providing a simple and efficient method for classification perfect for big datasets. Additionally, it facilitates the swift process even with a lot of images and videos. The low computational complexity makes it suitable for efficient classification. Thus, the integration of NB provides the ability to accurately extract the important characteristics in binary classifications.

2.4.2. Mutual information (MI)

In this research, identifying virtual montages, MI would be employed to measure the correspondence between different visual features (like color patterns, object presence, or temporal patterns in video) and media classification as a montage. With the application of methods such as the AFPO-MI-NB model, can employ MI to improve feature selection so that only the most informative features are employed for classification, with enhanced accuracy and efficiency in system.

MI theory, originally formulated to determine basic limits for compressing and communicating data reliably, employs entropy as a fundamental measure of information. Entropy measures the randomness of random variables and quantifies the amount of information exchanged between them efficiently. The idea is very helpful in areas that deal with examining relationships between variables, like feature selection in ML or data mining.

For simplicity, let's consider random variables that take discrete values. Let W and Z be two such random variables with Discrete values, where the uncertainty of W can be measures by its entropy $G(w)$ m, which is defined as Eq. (6).

$$G(w) = - \sum_{w \in W} o(w) \log(o(w)) \quad (6)$$

where, $o(w) = \Pr(W = w)$ is the probability density function of W . Note the entropy depends on the probability distribution of W , not its specific values. Similar, the joint entropy $W(W, Z)$ of W and Z is defined as Eq. (7).

$$W(W, Z) = - \sum_{w \in W} \sum_{z \in Z} o(w, z) \log(o(w, z)) \quad (7)$$

This measure the uncertainty in both W and Z when consider together, Conditional entropy, which represents the uncertainty of w given knowledge of Z is defined as Eq. (8).

$$G(W|Z) = \sum_{w \in W} \sum_{z \in Z} o(w, z) \log(o(w, z)) \quad (8)$$

where, $o(w, z)$ is the posterior probity of W given Z . If w is completely dependent on Z , then $G(W|Z) = 0$, implying no further uncertainty about W and Z is known. If knowing Z provide no information about W , then $G(W|Z) = G(W)$ The concept of MI quantifies how much information is shared between two variables W and Z . It defined as Eq. (9).

$$J(w; Z) = \sum_{w \in W} \sum_{z \in Z} o(w, z) \log \frac{o(w, z)}{o(w) o(z)} \quad (9)$$

To measures the reduction in uncertainty about W when Z is known, and vice versa. If W and Z are highly related, $J(W; Z)$ will be larger; if they are independent, $J(W; Z) = 0$. Additionally, MI can be rewritten as Eq. (10).

$$J(w; Z) = G(w) - G(W|Z) \quad (10)$$

This relation highlights that MI measures the amount of information about W that is shared with Z and it is equivalent to the reduction in the uncertainty of W when Z is known. Conditional MI: Similarly, Conditional MI, denoted as $J(W; Z|Y)$, measures the amount shared by W and Z when Y is known. It defined as $J(W; Z|Y) = G(W|Z) - G(W|Z, Y)$.

2.4.3. Feature weighting with adaptive flower pollination optimization (AFPO)

To improve on this, AFPO adapts feature weights dynamically in accordance with their relevance throughout the process of optimization, harmonizing global and local search strategies. Building on the flower pollination behavior, this technique improves feature selection and it enhances performance in processes. Such as virtual montage detection. Nevertheless, the computational complexity of AFPO, which varies with population size, generations, and feature space dimensions, is important for grasping scalability. The time complexity per iteration comes from

assessing feature relevance per iteration. Although AFPO performs optimally in exploring and exploiting, a deeper exploration of its time and space complexity is required to have a proper understanding of its computational efficiency, particularly with big data, it will have a comprehensive complexity analysis in the updated version to overcome this drawback and give a better insight into AFPO's efficiency. A population-based global optimization algorithm is known as an AFPO whose maintenance process is derived from the two pollination processes, such as cross and self-pollination between plant flowers, which are shown in Fig. 3, that act as the foundation for the optimization.

Following the regulations of *levy* transport, pollinators transport the petals of the plant's material along with the capacity to arbitrarily self-pollinate. The main advantage of the AFPO algorithm over other traditional swarm algorithms for optimization is its extraordinary ability for both regional growth and global research. The regional search is provided in Eq. (11).

$$w_j^{s+1} = w_j^s + v \times (w_i^s - w_l^s) \quad (11)$$

where w_i^s and w_l^s indicate the pollen outcome carrier. Values between 0 and 1 are picked by arbitrary character v . A Lévy movement has been demonstrated by insects that flew long distances in stages of different lengths. Consider that L is a Lévy distributed term provided in Eq. (12).

$$L = \frac{\lambda \Gamma(\lambda) \times \sin\left(\frac{\pi\lambda}{2}\right)}{\pi \times t^{1+\lambda}} \quad (12)$$

Where, the stage with $s > 0$ is the accurate dispersion, and $\Gamma(\lambda)$ is the Gamma function. The process of pollination intensity is expressed by the number of steps, which is comparable to a factor $\Gamma(\lambda)$. The cross-pollination is obtained using Eq. (13).

$$w_j^{s+1} = w_j^s + \gamma \times K(\lambda) \times (w_j^s - h^*) \quad (13)$$

where h^* is the suitable finding and s is the present looping. A scaling coefficient called γ is used to control the number of steps. In the AFPO process, an integer p is used as the affection chance for the switch from intensive localized pollinating to spontaneous global pollinating.

By improving its exploration-exploitation equilibrium AFPO improves feature selection from CNN-extracted features in the environment of virtual montage classification. This flexible technique makes improved performances of optimization in detecting related characteristics for effective montage classification, overcoming the issues in handling the unstructured data. Global pollination enables feature exploration in space by modeling distant communication pollination, while local pollination focuses on the local solutions in the neighbour-

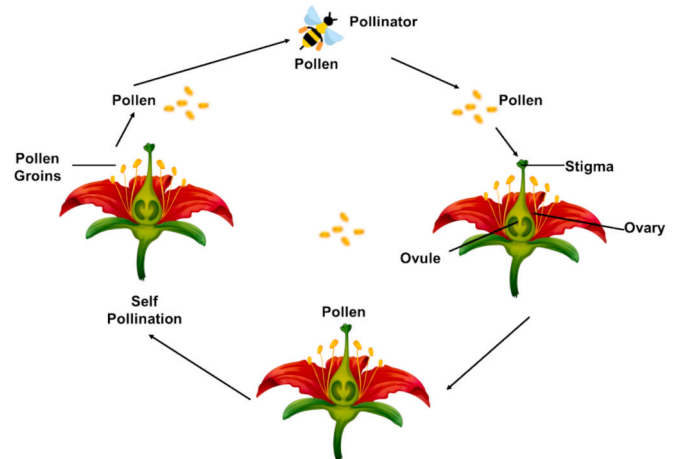


Fig. 3. AFPO pollinations.

hood. This adaptive switching mode enhances the AFPO performance to monitor large amounts of data. Eq. (14) demonstrates the adaptive probability for optimization.

$$p = p_{\min}(p_{\max} - p_{\min}) \times e^{-\alpha t} \quad (14)$$

where $p_{\max}$ is the maximum switching probability and $p_{\min}$ is the minimum switching probability. α Denotes the degradation ratio and t represents the execution frequency. The AFPO dynamically controls the p to switch between local and expandable region-searching processes based on the execution amount thereby enhancing the rate of speed convergence and efficiency in feature acquisition for virtual montage classification.

The AFPO-MI-NB hybrid model integrates to efficiently identify virtual montages from digital media. AFPO optimizes feature selection dynamically by weighting features at classification, contrary to conventional approaches based on fixed feature sets. MI computes the association between visual features (e.g., color patterns, object presence, temporal information) and the classification of montage, so that only the most informative features are employed. NB is utilized for effective classification by applying Bayes' probability theory to process binary classifications like montage vs. non-montage. AFPO integration NB performance by fine-tuning feature selection using global and local search strategies. This optimization enables the model to process unstructured data effectively and enhance classification precision. In total, the AFPO-MI-NB model provides a powerful and computationally effective solution to virtual montage detection in diverse media formats. Algorithm 1 provides the complete working process of the AFPO-MI-NB method.

Algorithm 1: AFPO-MI-NB

Step 1: Dataset Collection
Collect multimedia dataset (images/videos) from diverse sources.

Step 2: Preprocessing
For each data sample in the dataset:
Apply Min-Max Normalization:
 $y_{\text{new}} = (y - \min(y)) / (\max(y) - \min(y))$
Apply Data Augmentation:
– Apply transformations (flip, crop, rotate, color adjustments).
Remove low-quality samples.

Step 3: Feature Extraction using CNN
Initialize CNN:
For each layer:
Extract spatial features.
Perform dimensionality reduction to compact feature space.

Step 4: Naïve Bayes Classification
For each data sample:
Calculate probabilities for montage and non-montage:
 $d_NB = \arg\max(d.i \in D) P(d.i) * \prod [P(b.j | d.i)]$
Preliminary classification result (montage or non-montage).

Step 5: Mutual Information (MI) for Feature Selection
For each feature in the dataset:
Calculate MI between feature and target class:
 $J(W; Z) = \sum P(w, z) \log(P(w, z) / (P(w) * P(z)))$
Select features with high MI for better classification.

Step 6: AFPO Optimization for Feature Selection
Initialize AFPO parameters:
population, $p_{\min}$, $p_{\max}$, scaling factor (γ), execution frequency (t)
While not converged:
If local pollination:
Apply local pollination (Eq. 6)
Else:
Apply global pollination (Eq. 8) with Lévy flight.
Compute adaptive switching probability:
 $p = p_{\min} + (p_{\max} - p_{\min}) * e^{-\alpha t}$
Update population with optimized features.

Step 7: Reclassification Using Naïve Bayes on Optimized Features
For each data sample with optimized features:
Calculate probabilities for montage and non-montage:
 $d_NB = \arg\max(d.i \in D) P(d.i) * \prod [P(b.j | d.i)]$
Final classification result (montage or non-montage).

Step 8: Evaluate Model
Compute performance metrics (accuracy, precision, recall, etc.)
Return: Classified results and evaluation metrics.

3. Experimental results

This suggested method was verified using the dataset and consistency was ensured by quantifying the presentation using accuracy. This part narrates the experimental findings with the recommended AFPO-MI-NB method.

3.1. System configuration

The AFPO-MI-NB model was tested on the operating system of Windows 10 or 11 with an Intel core i9 processor, Memory of 32 GB, and Python 3.9 programming language, including libraries, such as Tensorflow and Keras for CNN implementation. Scikit learns for classification and optimization purposes. This configuration ensured robust performances and compatibility for ML tasks.

3.2. Evaluation metrics

Performance comparison in the section between the state-of-the-art approaches and the suggested method, Java implementation of C4.5 decision tree algorithm (J48) and Naïve Bayes [23]. Several metrics were measured as part of the assessment, including F1-measure, recall and precision.

Precision: It estimates the model's ability to correctly identify true montage instances across all instances classified as montages by reducing the incorrect affirmations. The following Eq. (15) is used for prediction calculation.

$$\text{precision} = \frac{TP}{TP + FP} \quad (15)$$

Recall: It is also known as the ability for responsiveness to evaluate the model's capacity to identify actual montages, while decreasing incorrect positivity, and it is evaluated by Eq. (16).

$$\text{recall} = \frac{TP}{FN + TP} \quad (16)$$

F1-measure: The equilibrium state of both recollection and accuracy provides a reasonable measurement in the situation of asymmetric class distributions. The Eq. (17) is used for calculating this concern.

$$F1 - \text{measure} = 2 * \frac{\text{recall} * \text{precision}}{\text{recall} + \text{precision}} \quad (17)$$

3.3. Comparison phase

This section provides detailed descriptions of the comparative analysis of the AFPO-MI-NB method with the conventional Naïve bayes method.

The Table 1 and Fig. 4 shows the precision values of various methods used in a particular task. J48 was a decision tree classifier that had a precision of 0.96 and Naïve Bayes was a probabilistic classifier with a precision of 0.93. The proposed method AFPO-MI-NB had a precision of 0.97 which is the highest precision score and the best performance of the methods evaluated in this task.

Table 2 and Fig. 5 displays the recall values from some of the methods tested on the task. The decision tree classifier J48 had a recall

Table 1
Outcomes of precision.

Methods	Precision
J48 [23]	0.96
Naïve bayes [23]	0.93
AFPO-MI-NB [Proposed]	0.97

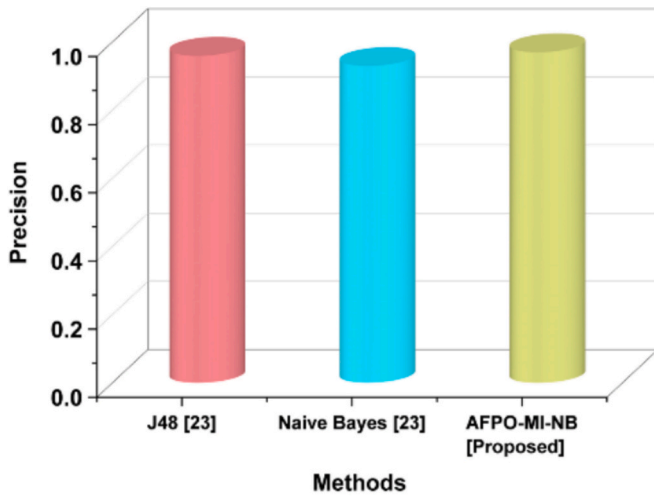


Fig. 4. Representation of precision.

Table 2
Numerical results of Recall.

Methods	Recall
J48 [23]	0.90
Naïve bayes [23]	0.84
AFPO-MI-NB [Proposed]	0.92

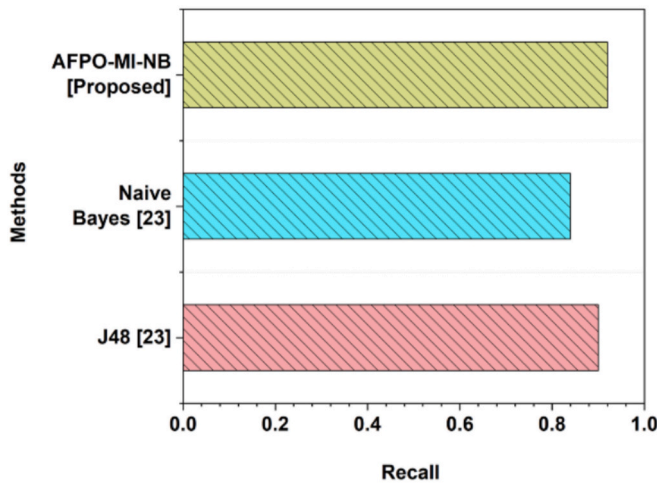


Fig. 5. Comparison of recall.

of 0.90 and Naïve Bayes on the other hand produced a lower recall of 0.84. The proposed AFPO-MI-NB method was better when looking at recall values, achieving a recall of 0.92 which shows that it demonstrated the ability to find relevant instances better than other methods that were tested.

Table 3 and Fig. 6 presents the F1-measure values for the different methods used in the task. J48, the decision tree classifier, produced an F1-score of 0.93, while Naïve Bayes produced a lower F1-score of 0.88.

Table 3
F1-measure outcomes.

Methods	F1-measure
J48 [23]	0.93
Naïve bayes [23]	0.88
AFPO-MI-NB [Proposed]	0.95

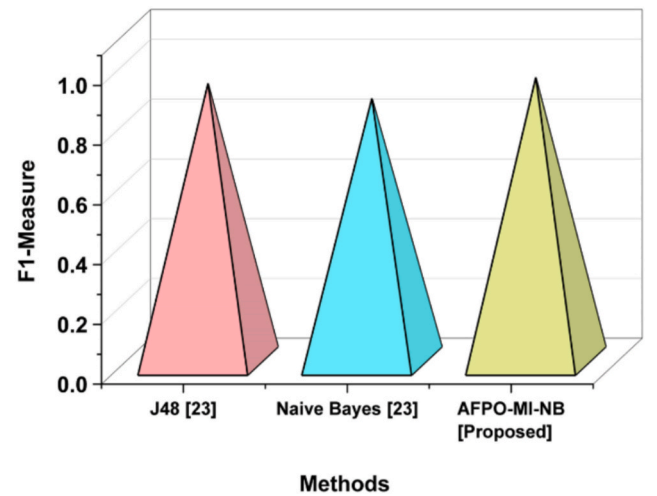


Fig. 6. Numerical performance of F1-measure.

Lastly, the proposed AFPO-MI-NB method produced an F1-score of 0.95 which is the highest score of all three methods and therefore provided the best overall F1-measure score for balancing both precision and recall.

3.4. Ablation research

The experimental results indicate that the complete AFPO-MI-NB model provides the best performance with F1 score = 0.95, recall = 0.92, precision = 0.97, and accuracy = 0.98. The performance drop when removing AFPO optimization shows its role in adjusting feature weights effectively. The performance drop when removing Mutual Information is noticeably larger and reflected the connection between relevant feature selection and correct identification of the virtual montage as shown in Table 4.

4. Discussion

The research offered an AFPO-MI-NB model as a perfect solution for classifying virtual montages across different media structures. The model can adapt to real-time information that has different styles, characteristics, and dimensions by using an extensive dataset that includes news, movies, social networking sites, and commercials. J48 [23] is susceptible to overfitting, particularly with noisy datasets. Similar risks exist with class imbalance where biased predictions may occur. Simple Naïve Bayes [23] classification algorithms can be accurate if the independence assumption holds true. Therefore, acceptance of the likelihood of dependence in features is essential as it will adversely affect the accuracy of the mapping. Any previous learning from

Table 4
Experiment of Ablation research.

Experiment	Experiment description	F1-Score	Recall	Precision	Accuracy
Full Model (AFPO-MI-NB)	Complete model with all components	0.95	0.92	0.97	0.98
Without AFPO Optimization	Model without AFPO for optimizing temporal features.	0.91	0.88	0.93	0.95
Without Mutual Information (MI)	Model without Mutual Information for selecting relevant features.	0.89	0.85	0.91	0.93

undisturbed datasets will no longer hold true. The implementation of the AFPO-MI-NB method will allow for AFPO selection to reduce over-fitted classifications and understand mapping, while pattern classification modes will also learn optimal feature selection for the models, and account for feature dependencies, that Naïve Bayes does not, allowing for better accuracy in predictions. Finally, under the circumstances with noisy and imbalanced data, the AFPO-MI-NB is more robust than J48 and Naïve Bayes. AFPO-MI-NB performs overall better when researching complex datasets where there are many types of methods to classify the data.

The work is shown to have real-world applicability in recognizing digital media, especially in industries like media and entertainment, security, and content authentication. For instance, in the media sector, the AFPO-MI-NB model can be applied to identify virtual montages within news videos or social media posts to verify content and counter fake news. In the security industry, it can help detect manipulated videos or images during forensic analysis, and maintain the integrity of digital evidence. It can also be used in the advertising sector to analyse and validate promotional material to avoid the propagation of false or manipulated media.

5. Conclusion

This research offered an innovative methodology, AFPO-MI-NB for identifying virtual montages in digital media, providing the present necessity for exact and effective approaches in visual information classifications. virtual montage detection dataset was collected and data pre-processing containing pixel normalization and data augmentation along with advanced feature extraction using CNN. The AFPO-MI-NB hybrid model is a strong and computationally efficient technique for identifying virtual montages from digital media. To dynamically optimizes feature selection using weighted features in classification, enabling more informative features. Mutual Information measures the relationship between visual features and montage classification, whereas Bayes' probability theory classifies binary ones. The model applies global and local search strategies to refine feature selection, improving the precision of classification and handling unstructured data efficiently. The MI-NB successfully notices the classification while the AFPO works the temporal feature selection to progress the model performance of accuracy and effectiveness in the model's ability to be appropriate across visual themes, content types, and media resolutions, and resolutions with the performance metrics demonstrating a greater F1-score of 0.95, recall of 0.92, and precision of 0.97. It attained a truthful resolution for areas like digital media analysis, copyright enforcement, and content control, providing a robust framework adaptable to varied media environments.

5.1. Limitations and future scope

Although the AFPO-MI-NB provides increased accuracy in montage identification, it may meet some limitations associated with CNN. A computational issue may arise when dealing with vast amounts of data. Additionally, it focuses on specific types of media while it generalizes for virtual reality (VR) and augmented reality (AR) applications. Future research should focus on improving the model's design or using light CNN changes for virtual montage identification. It could also explore emerging media types like VR and AR content and integrate with cloud-based services for efficient performance. The research has practical implications in sectors like media, entertainment, security, and content authentication. For instance, it can be used to identify virtual montages in news clips, forensic examinations, and advertising to prevent false or tampered media.

Declaration of competing interest

The author declares that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

There is yes funding.

Funding

(1) Research on the Construction of an Aesthetic Education System in Normal Universities Based on the Aesthetic Education Immersion Initiative, a general project of the 2024 Philosophy and Social Science Research Program for Jiangsu Colleges and Universities (Project Number: 2024SJYB0412).

(2) Research on the Interactivity between Public Aesthetic Education Courses and Professional Courses for Normal Students Based on the Aesthetic Education Immersion Initiative, a general project of the 2024 Special Program for Research on High-Quality Public Course Teaching Reform in Jiangsu Colleges and Universities (Project Number: 2024GZJX090).

Data availability

No data was used for the research described in the article.

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